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PENALTY METHODS FOR COMPUTING POINTS THAT SATISFY SECOND ORDER NECESSARY CONDITIONS

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Penalty methods using second derivatives are presented for computing points that satisfy second order necessary conditions for the constrained case in nonlinear programming. Convergence to such points is proved.

Key words: Constrained Optimization, Penalty Methods, Second Order Necessary Conditions.

1. Introduction

In mathematical programming the goal of almost all numerical methods is to compute points that satisfy first order necessary conditions. Only a few papers are devoted to computing points that satisfy second order necessary conditions; for the unconstrained case we can mention Zangwill and McCormick [1], McCormick [3], and for the constrained case: McCormick [4] where the algorithm presented is an extension of the variable reduction method.

The goal of this paper is to compute points that satisfy second order necessary conditions for constrained problems. In Section 1 preliminaries are presented which will be used throughout this paper. Section 2 contains a general convergence theorem. In Section 3 penalty techniques are given for solving our problem and convergence of these methods is proved with the use of the fundamental theorem.

1. Preliminaries

In this paper $\mathbf{R}^N$ is the usual vector space of real N-tuples with the usual inner product denoted by $(\cdot,\cdot)$, m a positive integer; $\langle 1, m\rangle$ denotes the set of integers included in the closed interval $[1, m]$. $f, f_i, i \in \langle 1, m\rangle$ are real-valued functions defined on $\mathbf{R}^N$ twice continuously differentiable; $\nabla f(x)$, $\nabla^2 f(x)$ denote the gradient and the Hessian matrix of f at x; $\nabla f(x)^t$ denotes the matrix row transpose of $\nabla f(x)$. Let C be the feasible set, that is:

$$C = \{x: f_i(x) \leq 0, \forall i \in \langle 1, m\rangle\}$$

C is assumed to be nonempty. For simplicity we shall restrict ourselves to inequality constraints only and consider the non-linear program

$$\mathrm{P}: \gamma = \inf(f(x) \mid x \in C).$$

Now we formulate a regularity condition which will be used for the convergence theorem:

Let x be any feasible point ($x \in C$) then the set of gradients $\nabla f_i(x)$, $i \in I(x)$ is linearly independent. (1.1)
($I(x) = \{i \in \langle 1, m \rangle : f_i(x) = 0\}$).

Under this condition, we can state the second-order necessary McCormick's theorem.

Theorem 1.1 [2]. *Necessary conditions that $\bar{x} \in C$ be a local minimum to problem* P *are that there exist real numbers λ_i, $i \in I(\bar{x})$ such that*

$$\lambda_i \geq 0, \quad i \in I(\bar{x}), \qquad \nabla f(\bar{x}) + \sum_{i \in I(\bar{x})} \lambda_i \nabla f_i(\bar{x}) = 0; \tag{1.2}$$

$$(y, \nabla f_i(\bar{x})) = 0, \quad \forall i \in I(\bar{x}) \Rightarrow \left(\left[\nabla^2 f(\bar{x}) + \sum_{i \in I(\bar{x})} \lambda_i \nabla^2 f_i(\bar{x}) \right] y, y \right) \geq 0. \tag{1.3}$$

The goal of this paper will be to compute points $\bar{x} \in C$ satisfying relations (1.2) and (1.3).

For the sequel we shall use concepts and theorems from the theory of point-to-set maps Ω defined from $\mathbf{R}^N$ to $\mathbf{R}^p$.

Ω is closed at a point $\bar{x}$ if $x_k \to \bar{x}$, $y_k \in \Omega(x_k)$ and $y_k \to \bar{y}$ imply that $\bar{y} \in \Omega(\bar{x})$.

Let $\{A_n\}$ be a sequence of subsets of $\mathbf{R}^N$. We define the outer limit, $\overline{\lim}_{n\to\infty} A_n$ by:

$$\overline{\lim_{n\to\infty}} A_n = \{x \in \mathbf{R}^N : x = \lim_{i\to\infty} x_{n_i}, \text{ where } \{n_i\} \text{ is an infinite subsequence of integers and } x_{n_i} \in A_{n_i}\}.$$

The inner limit $\underline{\lim}\, A_n$ by:

$$\underline{\lim_{n\to\infty}} A_n = \{x : x = \lim_{n\to\infty} x_n, \text{ where } x_n \in A_n \text{ for all but a finite number of } n\}.$$

If $\overline{\lim}_{n\to\infty} A_n = \underline{\lim}_{n\to\infty} A_n$ we say that the limit, $\lim_{n\to\infty} A_n$ exists and set

$$\lim_{n\to\infty} A_n = \overline{\lim_{n\to\infty}} A_n = \underline{\lim}\, A_n.$$

Let:

$$v(x) = \inf(\varphi(x, y) \mid y \in \Omega(x)),$$

where φ is a real-valued function defined on $\mathbf{R}^N \times \mathbf{R}^p$ and let

$$H(x) = \{y \in \Omega(x) \mid v(x) = \varphi(x, y)\}.$$

Theorem 1.2 (theorem 1.2.2 in [5]). *Suppose that for every sequence $\{x_n\}$ with $x_n \to x$, $\underline{\lim}\, \Omega(x_n)$ is either empty or equal to $\Omega(x)$ and that φ is continuous. Then H is closed at x.*

Theorem 1.3 (Corollary 2.3.4 in [5]). *Let g, $\{g_r\}$ be affine functions from $\mathbf{R}^N$ to $\mathbf{R}^p$ with $g = \lim_{r\to\infty} g_r$. If we define:*

$$H(g) = \{x: g(x) = 0\},$$

and suppose that

$$\limsup_{r\to\infty} \operatorname{rank}(g_r) \le \operatorname{rank}(g),$$

then either $\lim_{r\to\infty} H(g_r) = H(g)$ or $H(g_r)$ is empty for infinitely many r.

2. A fundamental convergence theorem

Theorem 2.1. *Let $\{x_n\}$ be a sequence converging to $x^* \in C$ and $\{\alpha_i^n\}$, $\{\lambda_i^n\}$ $(i \in \langle 1, m\rangle)$ real sequences such that:*

$$\lambda_i^n \ge 0, \quad \forall i \in \langle 1, m\rangle, \qquad \lim_{n\to\infty} \lambda_i^n = 0, \quad \forall i \notin I(x^*), \tag{2.1}$$

$$\lim_{n\to\infty} |\alpha_i^n| = 0, \quad \forall i \notin I(x^*). \tag{2.2}$$

Let $\{\varphi_i^n\}_{n\in\mathbf{N}}$ $(i \in I(x^*))$ be sequences of real-valued functions defined on $\mathbf{R}$ such that

$$\varphi_i^n(0) = 0, \quad \forall i \in I(x^*)\, \forall n. \tag{2.3}$$

If for every n, x_n satisfies

$$\nabla f(x_n) + \sum_{i=1}^m \lambda_i^n \nabla f_i(x_n) = 0 \tag{2.4}$$

and

$$\left(\left[\nabla^2 f(x_n) + \sum_{i=1}^m \lambda_i^n \nabla^2 f_i(x_n) + \sum_{i\notin I(x^*)} \alpha_i^n \nabla f_i(x_n) \nabla f_i(x_n)^{\mathrm{t}}\right] h, h\right)$$
$$+ \cdots + \sum_{i\in I(x^*)} \varphi_i^n((\nabla f_i(x_n), h)) \ge 0, \quad h \in \mathbf{R}^N, \tag{2.5}$$

then x^* satisfies (1.2) *and* (1.3).

Proof. (1) First we prove that $\{\lambda_i^n\}$ is a bounded sequence. We assume the contrary, and let $\lambda_n = \sum_{i=1}^m \lambda_i^n$; then we divide both sides of (2.4) by λ_n and take the limit as $n \to +\infty$. Then from (2.1) there exists u_i, $i \in I(x^*)$ such that

$$\text{(A)} \qquad u_i \geq 0, \qquad \sum_{i \in I(x^*)} u_i = 1,$$

$$\text{(B)} \qquad \sum_{i \in I(x^*)} u_i \nabla f_i(x^*) = 0.$$

From the regularity assumption it follows that $u_i = 0$, $\forall i \in I(x^*)$ and we obtain a contradiction with (A)

(2) Since the sequence $\{\lambda_i^n\}$ is bounded, relations (2.4) and (2.1) imply that there exists $\lambda_i \geq 0$, $i \in I(x^*)$, and a subsequence of $\{\lambda_i^n\}$ (we shall take without loss of generality the whole sequence) converging to λ_i such that

$$\nabla f(x^*) + \sum_{i \in I(x^*)} \lambda_i \nabla f_i(x^*) = 0$$

(3) We denote now:

$$D(x) = \{h : (\nabla f_i(x), h) = 0, \forall \mathrm{i} \in \mathrm{I}(\mathrm{x}^*)\}$$

Since $x^* = \lim_{n \to \infty} x_n$, since $0 \in D(x_n)$ for every n, by the regularity assumption we can apply Theorem 1.3 and obtain:

$$\text{(C)} \qquad D(x^*) = \lim_{n \to +\infty} D(x_n).$$

Let $h \in D(x^*)$ fixed and denote h_n the projection of h on $D(x_n)$. Since $0 \in D(x_n)$ and since the projection is a non expansive operator we have:

$$\|h_n\| = \|(h_n - 0)\| \leq \|(h - 0)\| = \|h\|.$$

So $\{h_n\}$ is bounded and thus there exists at least one cluster point of this sequence. Let h^* be an arbitrary cluster point of $\{h_n\}$. Using relation (C), we can apply Theorem 1.2 and since $h \in D(x^*)$ we obtain that $h^* = h$ and so we have proved that:

$$\text{(D)} \qquad h = \lim_{n \to +\infty} h_n.$$

From (2.5) we obtain with (2.3):

$$\text{(E)} \qquad \left(\left[\nabla^2 f(x_n) + \sum_{i \in I(x^*)} \lambda_i^n \nabla^2 f_i(x_n)\right] h_n, h_n\right) + \sum_{i \notin I(x^*)} \lambda_i^n (\nabla^2 f_i(x_n) h_n, h_n)$$

$$\sum_{i \notin I(x^*)} \alpha_i^n (\nabla f_i(x_n), h_n)^2 \geq 0.$$

Then we take the limit as $n \to +\infty$ in (E) and from relations (2.1), (2.2) and (D) we

obtain:

$$\text{(F)} \qquad \left(\left[\nabla^2 f(x^*) + \sum_{i \in I(x^*)} \lambda_i \nabla^2 f_i(x^*)\right] h, h\right) \geq 0.$$

Since h was taken arbitrary from $D(x^*)$ relation (F) is valid for every $h \in D(x^*)$ and Theorem 2.1 is proved.

3. Second order penalty methods

In [1, 3] for unconstrained problems second order descent methods were given for obtain points that satisfy second order necessary conditions. In this section we develop second order methods which consists of applying the former methods to exterior and interior penalty functions. For simplicity particular penalty functions are only used, but generalizations are trivial. For the sequel $\{r_n\}_{n \in \mathbf{N}}$, $\{K_n\}$ are sequences of positive real numbers such that:

$$\lim_{n \to \infty} r_n = 0, \tag{3.1}$$

$$\lim_{n \to \infty} K_n = +\infty, \qquad K_n \leq K_{n+1}, \quad \forall n \tag{3.2}$$

and z is a feasible point of C ($z \in C$).

3.1. Interior point techniques

We suppose in this subsection that f is inf-compact on C that is: for every λ the set $\{x \in C: f(x) \leq \lambda\}$ is compact. Then we assume that the set $C^0 = \{x: f_i(x) < 0, i \in \langle 1, m \rangle\}$ is non-empty and that $z \in C^0$. The classical interior point method consists of minimizing on C^0 the penalty function Ψ_n:

$$\Psi_n(x) = f(x) - r_n \sum_{i=1}^{m} \frac{1}{f_i(x)}, \tag{3.3}$$

but in fact, without convexity assumptions, only first necessary-points are obtained. Here we propose to apply for each n the second order descent methods given in [1, 3].

More precisely we construct (with the use of these methods) a sequence $\{x_n\}$ that satisfies:

$$x_n \in C^0, \qquad \Psi_n(x_n) \leq \Psi_n(z), \tag{3.4}$$

$$\nabla \Psi_n(x_n) = 0, \tag{3.5}$$

$$(\nabla^2 \Psi_n(x_n) h, h) \geq 0, \quad \forall h \in \mathbf{R}^N, \tag{3.6}$$

Remark. Ψ_n is twice continuously differentiable on C^0 and

$$\nabla \Psi_n(x) = \nabla f(x) + r_n \sum_{i=1}^{m} \frac{\nabla f_i(x)}{f_i(x)^2},$$

$$\nabla^2 \Psi_n(x) = \nabla^2 f(x) + r_n \sum_{i=1}^{m} \frac{\nabla^2 f_i(x)}{f_i(x)^2} - 2r_n \sum_{i=1}^{m} \frac{\nabla f_i(x) \cdot \nabla f_i(x)^t}{f_i(x)^3}.$$

Theorem 3.1. *Every cluster point of the sequence* $\{x_n\}$, *and there exists at least such a point, is a point of* C *that satisfies the second necessary order conditions* (1.2) *and* (1.3).

Proof. From (3.4) and the definition of Ψ_n we have:

$$f(x_n) \leq \Psi_n(x_n) \leq \Psi_n(z).$$

Thus from (3.1) there exists n_0, K such that

$$f(x_n) \leq K, \quad \forall n \geq n_0,$$

and since f is inf-compact on C the sequence $\{x_n\}$ is bounded. Let x^* be an arbitrary cluster point (such a point exists since $\{x_n\}$ is bounded), $x^* \in C$. Without loss of generality we can assume for simplicity that

$$x^* = \lim_{n \to \infty} x_n.$$

Now set

$$\lambda_i^n = \frac{r_n}{f_i(x_n)^2}, \qquad \alpha_i^n = \frac{-2r_n}{f_i(x_n)^3}, \qquad \varphi_i^n(t) = \frac{-2r_n}{f_i(x_n)^3} t^2.$$

Conditions (2.1)–(2.5) are obviously satisfied and we can apply Theorem 2.1.

3.2. Exterior point techniques

3.2.1. Second order method with the classical penalty function

The classical penalty method consists of minimizing on $\mathbf{R}^N$ the function

$$\Psi_n = f + K_n \sum_{i=1}^{m} (f_i^+)^2 \tag{3.7}$$

($a^+ = \max(a, 0)$). This function is not twice differentiable and the classical necessary second order conditions cannot be applied to such a function. Thus it is necessary as a first step to give necessary second order conditions. That is the goal of the next theorem.

Theorem 3.2. *Necessary conditions that* x *be a local minimum of* Ψ_n *on* $\mathbf{R}^N$ *are*:

$$\nabla \Psi_n(x) = 0 \tag{3.8}$$

and

$$\left(\left[\nabla^2 f(x) + 2K_n \sum_{i\in I^+(x)} f_i(x)\nabla^2 f_i(x)\right]h, h\right) + 2K_n \sum_{i\in I(x)} [(\nabla f_i(x), h)^+]^2$$
$$+ 2K_n \sum_{i\in I^+(x)} (\nabla f_i(x), h)^2 \geq 0, \quad \forall h \in \mathbf{R}^N, \tag{3.9}$$

where

$$I^+(x) = (i\colon f_i(x) > 0\}, \qquad \nabla \Psi_n(x) = \nabla f(x) + 2K_n \sum_{i=1}^{m} f_i^+(x)\nabla f_i(x),$$

Proof. (1) condition (3.8) is the usual order necessary condition.

(2) Consider for $h \in \mathbf{R}^N$, $t > 0$ the difference

$$\text{(A)} \qquad \Psi_n(x+th) - \Psi_n(x) = f(x+th) - f(x) + K_n \sum_{i=1}^{m} (f_i^+(x+th))^2 - f_i^+(x)^2.$$

There exists $\xi \in]x, x+th[$ such that

$$\text{(B)} \qquad f(x+th) - f(x) = t(\nabla f(x), h) + \frac{t^2}{2}(\nabla^2 f(\xi)h, h).$$

If $f_i(x) > 0$, for t sufficiently small there exist $\xi_i \in]x, x+th[$ such that:

$$\text{(C)} \qquad [f_i(x+th)^+]^2 - [f_i^+(x)]^2 = 2f_i(x)(\nabla f_i(x), h)$$
$$+ t^2([\nabla^2 f_i(\xi_i)f_i(\xi_i) + \nabla f_i(\xi_i)\cdot \nabla f_i(\xi_i)^t]h, h)$$

If $f_i(x) < 0$ for t sufficiently small we have:

$$\text{(D)} \qquad [f_i^+(x+th)]^2 - [f_i^+(x)]^2 = 0.$$

Consider now the case where $f_i(x) = 0$

(a) If $(\nabla f_i(x), h) < 0$ then for t sufficiently small, we have

$$\text{(E)} \qquad [f_i^+(x+th)]^2 - [f_i^+(x)]^2 = 0.$$

(b) If $(\nabla f_i(x), h) > 0$ then for t sufficiently small there exists $\eta_i \in]x, x+th[$ such that:

$$\text{(F)} \qquad [f_i^+(x+th)]^2 - [f_i^+(x)]^2 = t^2([f_i(\eta_i)\nabla^2 f_i(\eta_i) + \nabla f_i(\eta_i)\nabla f_i(\eta_i)^t]h, h).$$

(c) If $(\nabla f_i(x), h) = 0$ let t fixed. From Taylor's theorem there exists $\eta_i \in]x, x+th[$ such that:

$$\text{(F}'\text{)} \qquad [f_i(x+th)]^2 - [f_i(x)]^2 = t^2([f_i(\eta_i)\nabla^2 f_i(\eta_i) + \nabla f_i(\eta_i)\nabla f_i(\eta_i)^t]h, h).$$

Set $\beta_i(x, t, h) = [f_i(x+th)]^2 - [f_i(x)]^2$. Then since $f_i(x) = f_i^+(x) = 0$ and since:

$$\text{(F}''\text{)} \qquad [f_i^+(x+th)]^2 - [f_i^+(x)]^2 = \beta_i(x, t, h) \quad \text{if } f_i(x+th) > 0,$$
$$= 0 \qquad \text{if } f_i(x+th) \leq 0,$$

if we replace in (F″) β_i by its expression in (F′) we obtain both possibilities (E)

and (F) so that in all cases we have:

$$\text{(G)} \qquad i \in I(x) \Rightarrow \lim_{t \to 0} \frac{[f_i^+(x+th)]^2 - [f_i^+(x)]^2}{t^2} = [(\nabla f_i(x), h)^+]^2.$$

From relations (3.8), (B), (C), (D), (E), (F), (G) we obtain:

$$\text{(H)} \qquad \frac{\Psi_n(x+th) - \Psi_n(x)}{t^2} = \tfrac{1}{2}(\nabla^2 f(\xi)h, h) + K_n\Big[\sum_{i \in I^+(x)} ([\nabla^2 f_i(\xi_i) f_i(\xi_i) + \nabla f_i(\xi_i) \cdot \nabla f_i(\xi_i)^t]h, h) + \sum_{i \in I(x)} \alpha_i(x, t, h)\Big],$$

where $\alpha_i(x, t, h) \to [(\nabla f_i(x), h)^+]^2$ if $t \to 0^+$. Then if we take the limit in (H) when $t \to 0^+$, we obtain (3.9) since

$$\lim_{t \to 0^+} \frac{\Psi_n(x+th) - \Psi_n(x)}{t^2} \geq 0.$$

Now we can give a second order method with the penalty function Ψ_n defined by (3.7). Such a method consists to construct a sequence $\{x_n\}$ which satisfies the second order necessary condition that is

$$\nabla \Psi_n(x_n) = 0, \tag{3.10}$$

$$\left(\left[\nabla^2 f(x_n) + 2K_n \sum_{i \in I^+(x_n)} f_i(x_n) \nabla^2 f_i(x_n)\right] h, h\right) + 2K_n \sum_{i \in I(x_n)} [(\nabla f_i(x_n), h)^+]^2 + 2K_n \sum_{i \in I^+(x_n)} (\nabla f_i(x_n), h)^2 \geq 0, \quad \forall h \in \mathbf{R}^N. \tag{3.11}$$

For n fixed we suppose that x_n can be obtained by a descent method such that we have:

$$\Psi_n(x_n) \leq \Psi_n(z) \tag{3.12}$$

and we assume that Ψ_0 is inf-compact that is: for every λ the set $\{x\colon \Psi_0(x) \leq \lambda\}$ is compact.

Theorem 3.3. *The same as Theorem* 3.1.

Proof. (1) Let x^* be an arbitrary cluster point of the sequence $\{x_n\}$. From (3.12), (3.2) and since $z \in C$ we have:

$$\Psi_0(x_n) \leq f(z)$$

such that $\{x_n\}$ is bounded and we can assume without loss of generality that

$x^* = \lim_{n\to\infty} x_n$. From (3.12) we have:

$$[f_i^+(x^*)]^2 \le \lim_{n\to\infty} \frac{f(z)-f(x^*)}{K_n} = 0$$

so that x^* belongs to C.

(2) Since $x_n \to x^* \in C$ there exists $n_0 > 0$ such for $n \ge n_0$:

$$I(x_n) \subset I(x^*), \qquad I^+(x_n) \subset I(x^*).$$

Then set:

$$\begin{aligned} \lambda_i^n &= 2K_n f_i^+(x_n), \qquad \alpha_i^n = 0, \quad i \in \langle 1, m\rangle, \\ \varphi_i^n(t) &= 2K_n t_+^2 \quad \text{if } i \in I(x_n), \\ &= 2K_n t^2 \quad \text{if } i \in I^+(x_n), \\ &= 0 \qquad\quad \text{if } i \notin I(x_n) \cup I^+(x_n). \end{aligned}$$

Conditions (2.1)–(2.5) are obviously satisfied and we can apply Theorem 2.1.

3.2.2. Exterior techniques with twice continuously differentiable functions

The former method supposed that one can construct a sequence $\{x_n\}$ satisfying relations (3.10), (3.11) and (3.12). Methods given in [1, 3] cannot be applied to find such points. Thus this problem is open.

Now we give an exterior technique with a twice continuously differentiable function for which the methods given in [1, 3] can be applied. The penalty function is the following:

$$\Psi_n(x) = f(x) + K_n \sum_{i=1}^{m} f_i^+(x)^3.$$

and we construct a sequence $\{x_n\}$ satisfying:

$$\Psi_n(x_n) \le \Psi_n(z), \tag{3.13}$$

$$\nabla \Psi_n(x_n) = 0, \tag{3.14}$$

$$(\nabla^2 \Psi_n(x_n)h, h) \ge 0, \quad \forall h \in \mathbf{R}^N. \tag{3.15}$$

Here:

$$\nabla \Psi_n(x) = \nabla f(x) + 3K_n \sum_{i=1}^{m} [f_i^+(x)]^2 \nabla f_i(x)$$

$$\nabla^2 \Psi_n(x) = \nabla^2 f(x) + 3K_n \sum_{i=1}^{m} [2f_i^+(x)\nabla f_i(x) \cdot \nabla f_i(x)^t + f_i^+(x)^2 \nabla^2 f_i(x)]$$

and we assume that Ψ_0 is inf-compact on $\mathbf{R}^N$.

Theorem 3.4. *Same as Theorem* 3.2.

Proof. Let x^* be a cluster point of sequence $\{x_n\}$. As in the proof of Theorem 3.3

such a point exists, belongs to C and we can suppose that

$$x^* = \lim_{n\to\infty} x_n.$$

Now let

$$\lambda_i^n = 3K_n f_i^+(x_n)^2, \qquad \alpha_i^n = 0, \quad \forall i \in \langle 1, m\rangle,$$
$$\varphi_i^n(t) = 6K_n f_i^+(x_n)t^2, \quad \forall i \in I(x^*).$$

Since $x_n \to x^* \in C$ there exists $n_0 > 0$ such that for $n \geq n_0$:

$$I(x_n) \subset I(x^*), \qquad I^+(x_n) \subset I(x^*).$$

and then conditions (2.1)–(2.5) are obviously satisfied.

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